Probability and Statistics
Topic 5. Expectation

- Expected value
- Properties of expectation
- Variance and moments
- Properties of variance
- Covariance and correlation
- Probability generating function
- Properties of p.g.f.

Seminar 5

Problem 1. If $\mathsf{E}(X) = 1$, $\mathsf{E}(Y) = -1$, $\mathsf{Var}(X) = 4$, and $\mathsf{Var}(Y) = 3$, find

(a) $\mathsf{E}\big((X-1)\ln 2\big)$; (b) $\mathsf{E}(3X^2 - 5X)$; (c) $\mathsf{E}((-4-Y)Y)$; (d) $\mathsf{Var}((\frac{1}{2} - X)X)$.

If $\mathsf{E}(X) = 2$, $\mathsf{E}(Y) = -1$, $\mathsf{E}(X^2) = 5$, and $\mathsf{E}(Y^2) = 4$, find

(e) $\mathsf{E}\big(X - eY + e^2\big)$; (f) $\mathsf{Var}(\pi + \sqrt{3}Y)$; (g) $\mathsf{E}((5-X)Y)$; (h) $\mathsf{Var}(\frac{1}{2}X + \frac{1}{3}Y)$.

Problem 2. An urn contains 10 balls numbered 1 to 10. Two balls are sampled in turn without replacement and their numbers are denoted as X_1 and X_2 respectively. Let Y be the minimum of X_1 and X_2.

(a) Show that the distributions of X_1 and X_2 coincide.

(b) Find expectation and variance of X_1.

(c) Find covariance of X_1 and X_2. Can you guess the sign before calculations? Are X_1 and X_2 independent?

(d) Find expectation and variance of Y.

(e) Find the conditional distribution of X_1 given that $Y = y$.
Hint: the case $X_1 = y$ is special

(f) Find expectation and variance of X_1 given that $Y = y$.

Problem 3. A fair coin is tossed repeatedly and independently until two consecutive heads or two consecutive tails appear. Denote by N the r.v. equal to the number of tosses made.

(a) Find the p.m.f. of the r.v. N.

(b) Find the expected value of the r.v. N.

(c) Find the variance of the r.v. N.

Assume now that the coin is tossed until we obtain a tail that is immediately preceded by a head; let a r.v. N_1 denote the corresponding number of tosses.

(d) Find the p.m.f. and the expected value of the r.v. N_1.

Problem 4. A die with 6 faces, numbered $1, \ldots, 6$, is rolled a fixed number of times n. The probability that the ith face comes up on each roll is denoted p_i and the results of different rolls are assumed independent. Let X_i be the number of times that the ith face comes up.

(a) Find the joint p.m.f. $p_{X_1,\ldots X_r}(k_1, \ldots, k_r) = \mathsf{P}(X_1 = k_1, \ldots, X_r = k_r)$ for $r = 1, 2$, and 6.

(b) Find the expected value of X_i.

(c) Find the variance of X_i.

(d) Find $E[X_i X_j]$ for $i \neq j$.

Problem 5. Assume that X is a **r.v.** with geometric distribution $\mathscr{G}(p)$.

(a) Calculate the *probability generating function* (p.g.f.) G_X of X.

(b) Use G_X to find $\mathsf{E}(X)$, the second factorial moment $\mu^{(2)}(X)$ and $\mathsf{Var}(X)$

Assume further that X_1, X_2, ... are r.v.'s with geometric distribution $\mathscr{G}(p)$ and N is a r.v. with Poisson distribution $\mathscr{P}(\lambda)$. Assume that the r.v.'s $N, X_1, X_2, \ldots,$ are independent and set $S := X_1 + \cdots + X_N$.

(c) Calculate the p.g.f. G_S of the r.v. S.(Hint: condition on the event $\{N = n\}$, $n \in \mathbb{Z}_+$)

Problem 6. In the morning, the centipede put shoes on all $2m$ of its legs. During the day, each shoe could be lost with probability p independently of the others. What is the expected number of pairs of legs in shoes in the evening if there are n shoes left? Hint: calculate the expectation of the indicator I_k that both shoes in the pair no. k are not lost,conditioned on the number of soes left $N = n$.

Problem 7. The children's library has 4 shelves with books. There are 35 books on the first shelf, 20 on the second, 45 on the third, and 60 on the fourth. Alice randomly chooses one book. Let X denote the number of books on the shelf from which Alice chose the book. Bob first randomly selects the shelf from which he will later choose a book. Let Y denote the number of books on the shelf chosen by Bob.

(a) Which of $\mathsf{E}(X)$ anf $\mathsf{E}(Y)$ do you think is larger? Give your reasoning in words.

(b) Compute $\mathsf{E}(X)$ and $\mathsf{E}(Y)$.

Problem 8. A fair die is rolled. How many rolls do you expect to do before you observe every of faces of a die at least once?

Extra problems

(to be discussed if time permits)

Problem 9. Two balls are chosen randomly from an urn containing 8 white, 4 black, and 2 orange balls. Suppose that we win \$2 for each black ball selected and we lose \$1 for each white ball selected. Let X denote our winnings. Find the expected value and variance of the random variable X.

Problem 10. Suppose that a fair die is rolled twice. Find *expectation* and *variance* for the following random variables:

(a) the maximum value to appear in the two rolls;

(b) the minimum value to appear in the two rolls;

(c) the value of the first roll minus the value of the second roll.

Problem 11. Assume X and Y have joint probability mass function

$p(x,y)$	$y=-1$	$y=1$	$y=2$	$y=3$
$x=0$	$1/12$	$1/4$	0	0
$x=1$	$1/12$	$1/12$	$1/12$	$1/12$
$x=3$	0	0	$1/4$	$1/12$

(a) Find $\mathsf{E}(X)$, $\mathsf{E}(Y)$, $\mathsf{E}(S)$, $\mathsf{E}(D)$, and $\mathsf{E}(M)$, where $S = X+Y$, $D = X-Y$, and $M = \max\{X,Y\}$.

(b) Find whether X is independent of Y.

(c) Find variance of the random variables S and M.

(d) Find covariance of X and D.

(e) Find $\mathsf{E}(Y \mid X = x)$ for $x = 0, 1, 3$.